

Analysis of magnetohydrodynamics oscillations at Heliotron J with Python

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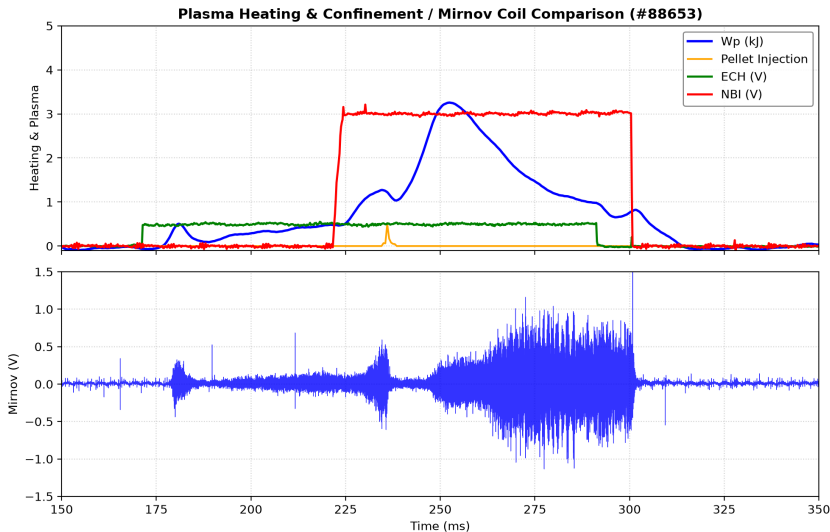
Institute of Advanced Energy, Kyoto University

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Plasma Heating and Confinement vs. Mirnov Signal



Objective 1

Dominant Mode Identification

- Identify the dominant coherent MHD mode(s) present in the Mirnov coil array using a dual criterion: amplitude above a 3σ noise threshold *and* spatial coherence $\gamma^2 > 0.7$ across at least one coil pair.
- Determine the toroidal mode number n and localize the fluctuation radially using the ECE and PMP diagnostics.
- Repeat the analysis separately for the ECH+NBI and NBI-only phases of shot #88653, to check whether mode identity and localization survive the loss of ECH heating.

Objective 1: Signal Processing Parameters

Time-Frequency Analysis Settings (Shot #88653)

- **FFT Spectrogram:** Hann window applied, segment length $T = 2.00$ ms, achieving a frequency resolution of $\Delta f = 500.0$ Hz.
- **Wavelet Transform:** Morlet continuous wavelet transform (CWT) evaluated across 150 frequencies (10 kHz – 150 kHz).
- **Noise Baseline:** Calibrated between 150 ms and 175 ms.
- **Validation:** Dominant modes must exceed a 3σ threshold above the baseline and demonstrate spatial coherence ($\gamma^2 > 0.7$) across at least one coil pair.

Objective 1: Dominant Mode Identification

Confirmed Dominant and Coherent Modes (Shot #88653)

- 89.0 kHz **Mode:** Passed dual-criterion test with robust spatial coherence across all pairs (MP1-MP3: 0.855, MP1-MP4: 0.876, MP3-MP4: 0.878).
- 95.5 kHz **Mode:** Passed dual-criterion test, confirmed by MP3-MP4 coherence (0.710).
- **Observation:** Both modes initiate sharply just after 175 ms, consistent with a rapid MHD onset during the discharge.

Objective 1: Justification for a Single Primary Mode

Evidence for a Single High-Frequency Instability

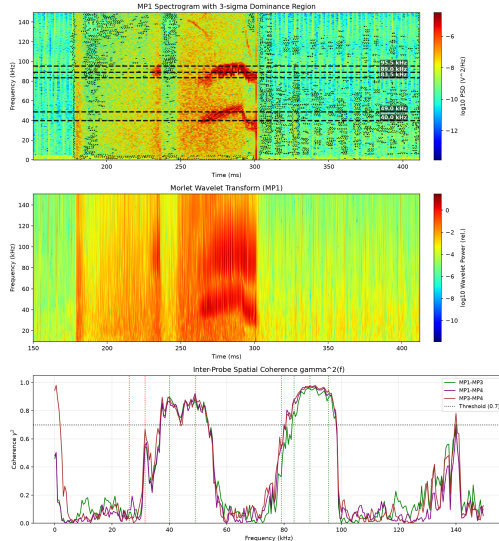
- **Unified poloidal structure:** the 80.0 kHz–120.0 kHz band contains only one dominant poloidal mode ($m = +3$ at 89.84 kHz).
- **Frequency sweeping (chirping):** the analysis flags a frequency-sweep interval and isolates a flat-frequency sub-window for further processing. The apparent 95.5 kHz detection is most likely the spectral footprint of the 89.0 kHz mode drifting in frequency during the discharge, rather than a separate instability.
- **Sensitivity to the search boundary:** closely spaced spectral peaks can appear or disappear depending on how the search window is defined, so such detections are not necessarily distinct physical modes.
- On this basis, the 89.0 kHz signal is taken as the primary mode for the $n = -5$ toroidal fit and radial localization that follow.

Objective 1: Spatial and Radial Localization

Structural Profile of the 89.0 kHz Target Mode

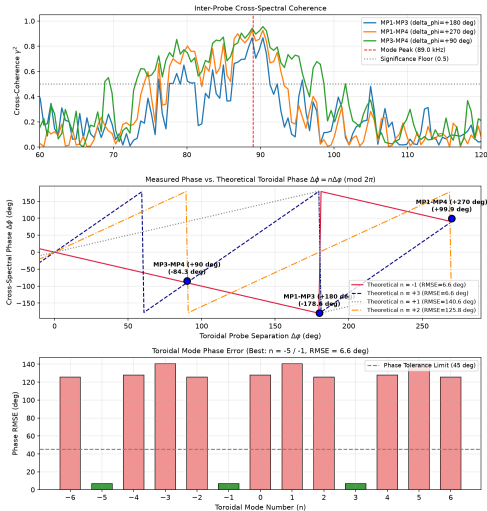
- The best-fit toroidal mode number was identified as $n = -5$ (aliased as $n = -1$) with a minimum Root Mean Square Error (RMSE) of 6.6° .
- Radial fluctuation amplitudes peaked near the plasma edge/core boundary on the ECE13FAST channel (66.5 GHz) reaching a Root Mean Square (RMS) of 0.478 V.
- Maximum radial coherence was observed deeper within the plasma on the ECE03FAST channel (61.5 GHz) with $\gamma^2 = 0.187$.
- Edge fluctuation coupling was confirmed dominantly at HAFast7.5 ($\gamma^2 = 0.2335$, RMS = 1.25×10^{-4} V), demonstrating core-edge mode linkage.

Objective 1: Spectrogram and Spatial Coherence



Objective 1: Primary Mode Toroidal Identification

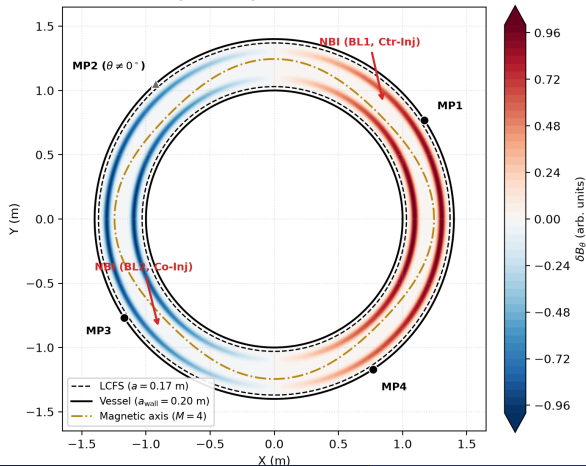
Primary Mode Toroidal Identification - Shot 88653
(f ~ 89.0 kHz, Window 259.1-275.0 ms)



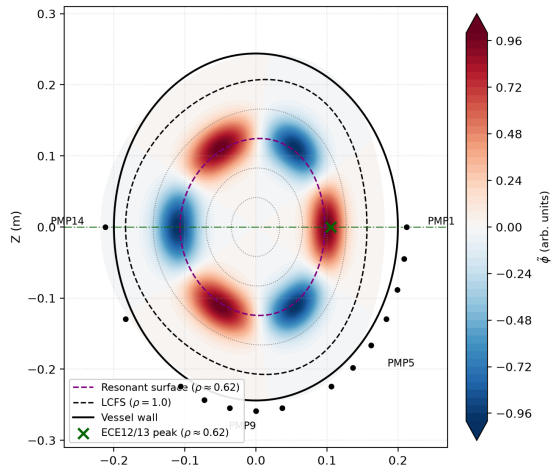
Objective 1: Primary Mode 2D Spatial Structure

Heliotron J #88653 — Primary Mode 2D Spatial Structure ($f \approx 89.0$ kHz, $m = 3$, $n = -1$)

(a) Top-down Equatorial View ($n = -1$)

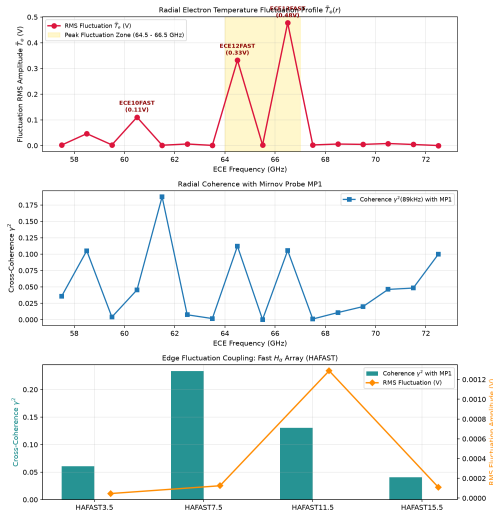


(b) Poloidal Cross-Section ($m = 3$)



Objective 1: Radial Fluctuation Profile

Primary Mode Radial Fluctuation & Localization Profile - Shot 88653
($f \sim 89.0$ kHz, Window 259.1-275.0 ms)



Objective 1: Comparative Analysis — ECH+NBI vs. NBI-only

Extending Mode Identification Across Heating Regimes (Shot #88653)

- ECH is disconnected at $t = 291.300$ ms. Two windows are compared: **ECH+NBI** (259.1 ms to 291.3 ms, 32.2 ms duration) and **NBI-only** (291.3 ms to 300.0 ms, 8.7 ms duration).
- *Caveat:* the NBI-only window is roughly $3.7\times$ shorter, so it yields fewer independent segments for statistical estimators — absolute magnitudes should be compared with this in mind.
- The same dual-criterion (3σ amplitude + $\gamma^2 > 0.7$) pipeline is re-applied independently to each window.

Objective 1 Comparative: Continuous vs. Bursty Dynamics

ECH+NBI (259.1–291.3 ms)

- Primary (70–110 kHz): 87.60(533) kHz, range 74.2–97.7 kHz
- Secondary (30–65 kHz): 45.47(585) kHz, range 30.3–64.5 kHz
- 3σ occupancy: 100.0% (both modes)
- Jump rate: 1.06 (primary) / 1.27 (secondary) jumps/ms
- Coexisting branches: secondary only, at 40.5 & 51.5 kHz

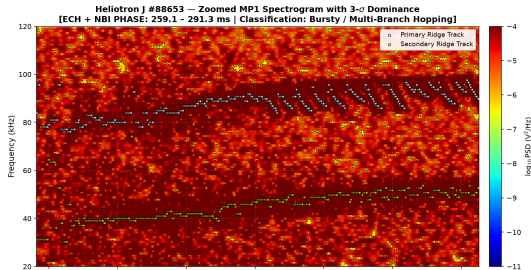
NBI-only (291.3–300.0 ms)

- Primary (70–110 kHz): 86.69(302) kHz, range 81.1–95.7 kHz
- Secondary (30–65 kHz): 40.26(427) kHz, range 35.2–50.8 kHz
- 3σ occupancy: 100.0% (both modes)
- Jump rate: 1.26 (primary) / 0.46 (secondary) jumps/ms
- Coexisting branches: primary at 86.0 kHz; secondary at 37.3 kHz

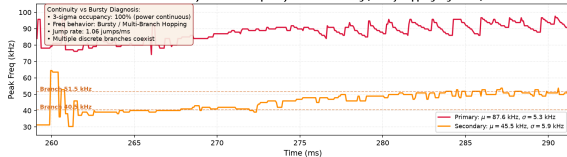
Classification: Bursty / multi-branch hopping in both windows – mode energy is continuously sustained by the NBI fast-ion drive rather than decaying, but the instantaneous frequency hops discretely between 2–3 branches instead of drifting smoothly.

Objective 1 Comparative: Zoomed Spectrograms (3σ Dominance)

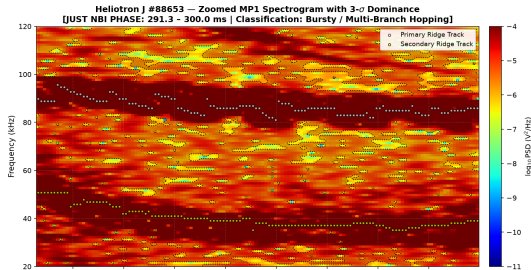
ECH+NBI (259.1–291.3 ms)



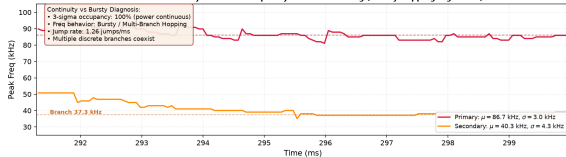
Tracked Mode Trajectories & Frequency Branch Switching (Bursty Hopping Signature)



NBI-only (291.3–300.0 ms)



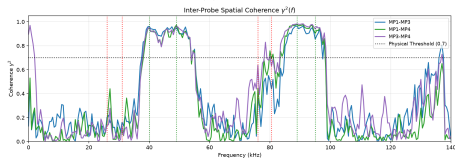
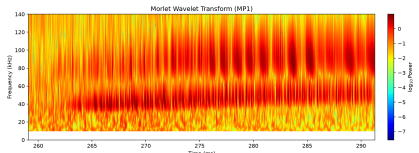
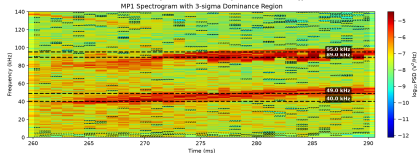
Tracked Mode Trajectories & Frequency Branch Switching (Bursty Hopping Signature)



Objective 1 Comparative: Mode Identification Overview

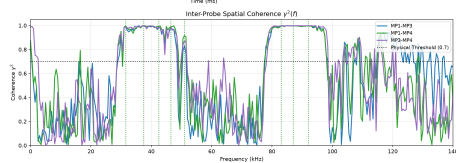
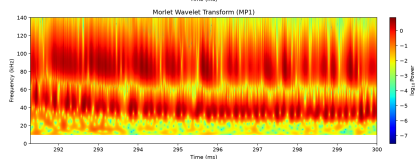
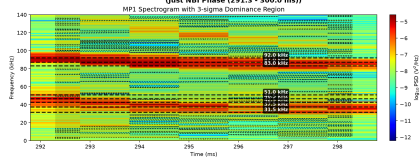
ECH+NBI

Heliotron J #88653 — Objective 1 Mode Identification
(ECH + NBI Phase (259.1 - 291.3 ms))



NBI-only

Heliotron J #88653 — Objective 1 Mode Identification
(Just NBI Phase (291.3 - 300.0 ms))



Objective 1 Comparative: Confirmed Discrete Modes

ECH+NBI

f (kHz)	Amp. (a.u.)	γ_{13}^2	γ_{14}^2	γ_{34}^2
89.0	5.36×10^{-6}	0.962	0.975	0.983
49.0	3.05×10^{-6}	0.950	0.972	0.964
40.0	2.89×10^{-6}	0.962	0.952	0.953
95.0	1.78×10^{-6}	0.854	0.904	0.939

4 additional candidates (80.5, 26.0, 76.0, 31.0 kHz) were above the noise floor but failed the coherence criterion.

NBI-only

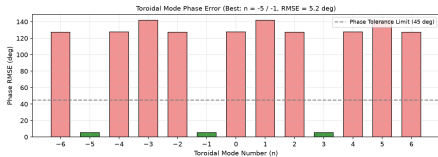
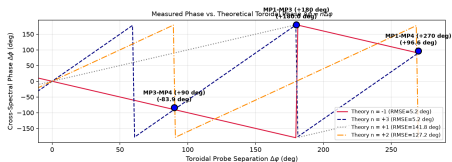
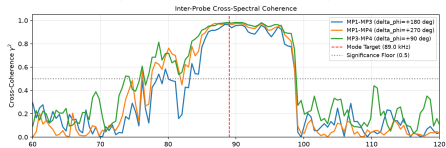
f (kHz)	Amp. (a.u.)	γ_{13}^2	γ_{14}^2	γ_{34}^2
37.5	8.24×10^{-6}	0.997	0.997	0.999
87.0	6.76×10^{-6}	0.998	0.996	0.997
83.0	3.90×10^{-6}	0.996	0.996	0.998
92.0	3.43×10^{-6}	0.998	0.997	0.999
42.5	2.41×10^{-6}	0.984	0.995	0.989
31.5	1.85×10^{-6}	0.990	0.997	0.989
46.5	1.21×10^{-6}	0.981	0.988	0.982
51.0	2.17×10^{-7}	0.829	0.916	0.872

Observation: every candidate mode passes the dual criterion once ECH switches off, and coherence pairs are systematically higher ($\gamma^2 \gtrsim 0.98$ vs. ~ 0.85 – 0.98) – consistent with a cleaner, more coherent fluctuation spectrum in the NBI-only phase.

Objective 1 Comparative: Toroidal Mode Identification

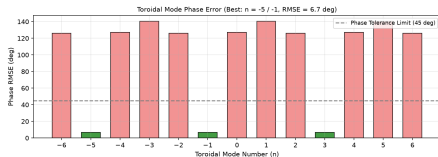
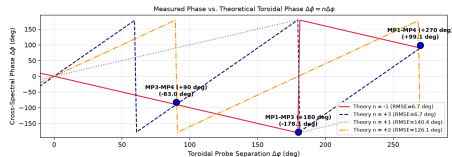
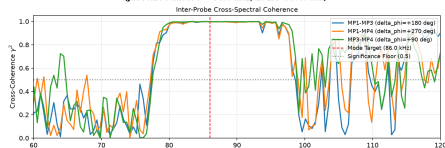
ECH+NBI

Primary Mode Toroidal Identification — Shot 88653
(ECH + NBI Phase: $f \sim 89.0$ kHz, 259.1-291.3 ms)



NBI-only

Primary Mode Toroidal Identification — Shot 88653
(Just NBI Phase: $f \sim 86.0$ kHz, 291.3-300.0 ms)



Objective 1 Comparative: Radial Fluctuation Profile

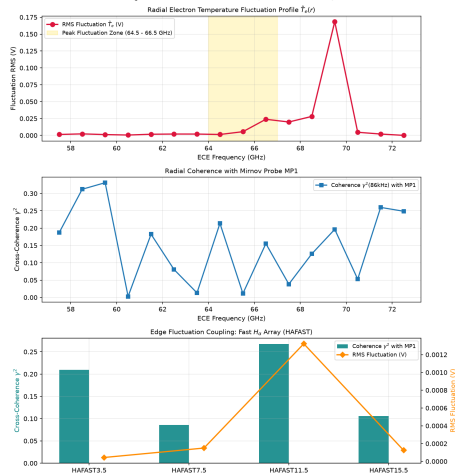
ECH+NBI

Primary Mode Radial Fluctuation & Edge Coupling — Shot 88653
(ECH + NBI Phase: $f \sim 89.0$ kHz, 259.1-291.3 ms)



NBI-only

Primary Mode Radial Fluctuation & Edge Coupling — Shot 88653
(Just NBI Phase: $f \sim 86.0$ kHz, 291.3-300.0 ms)



Objective 1 Comparative: Toroidal Fit Persists Across Regimes

- **ECH+NBI** (target 89.0 kHz): best fit $n = -5$ (aliased $n = -1$), RMSE = 5.2° ; pair coherences $\gamma^2 = 0.962$ (MP1-MP3), 0.974 (MP1-MP4), 0.982 (MP3-MP4).
- **NBI-only** (target 86.0 kHz): best fit $n = -5$ (aliased $n = -1$), RMSE = 6.7° ; pair coherences $\gamma^2 = 0.997$, 0.997, 0.998.
- **Physical reading:** the primary mode retains the *same* toroidal mode number and helicity across the ECH shut-off, even though its central frequency shifts – consistent with the same underlying instability persisting under a purely NBI-driven fast-ion population.

Objective 2

Spatial Structure & Dynamics

- Separate MHD activity into a low-frequency band (40 kHz to 80 kHz) and a high-frequency band (80 kHz to 120 kHz), and characterize the dominant poloidal mode number in each.
- Test candidate physical scaling models (current-driven, Alfvénic, BAE/GAM) against the observed correlation behavior of each band.
- Quantify non-linear self-coupling between the fluctuation carrier and its own envelope.
- Compare spatial structure, envelope dynamics, and macroscopic correlations between the ECH+NBI and NBI-only phases.

Objective 2: Dual-Band Dynamic Structure

Separation of MHD Phenomena (Shot #88653)

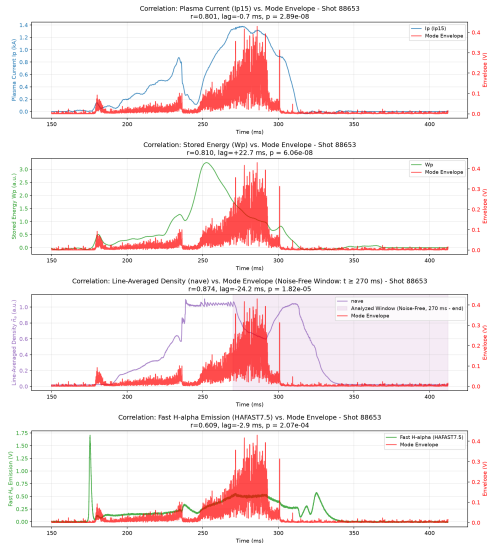
- **Low-Frequency Band (40.0 kHz – 80.0 kHz):**

- Dominant poloidal mode identified as $m = +2$ at 42.97 kHz.
- The optimal physical model fit is the "Current-induced (I_p)" model, yielding a strong correlation of $|r| = 0.8465$.
- Theoretical Alfvén scaling validation returned a negative correlation ($r = -0.439$).

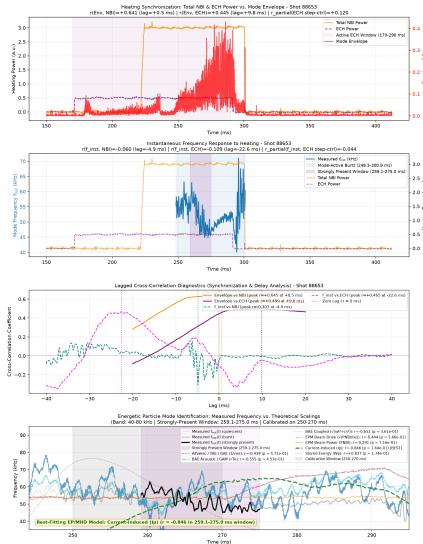
- **High-Frequency Band (80.0 kHz – 120.0 kHz):**

- Dominant poloidal mode identified as $m = +3$ at 89.84 kHz.
- Best-fitting energetic particle/MHD behavior aligns with the "BAE Acoustic / GAM ($\sqrt{T_e}$)" model ($|r| = 0.4789$).
- Alfvén scaling validation remains weak ($r = 0.161$).

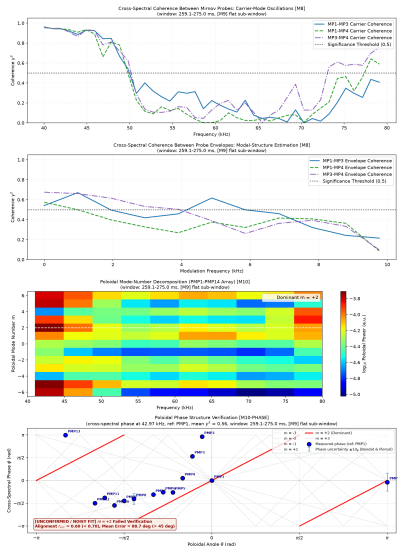
Objective 2: LF Band Macroscopic Correlations



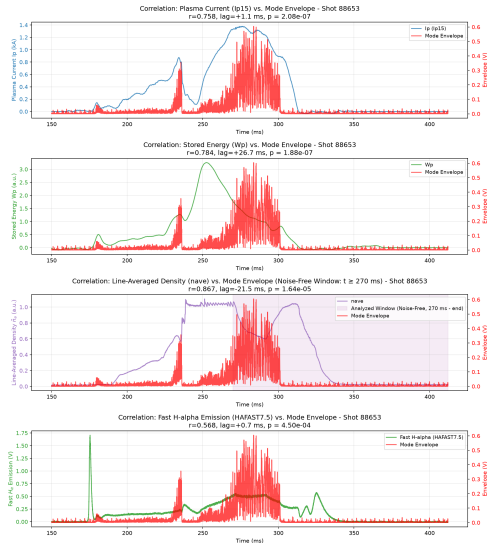
Objective 2: LF Band Heating Synchronization



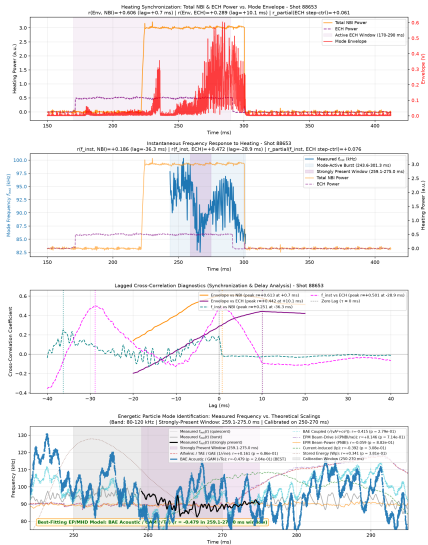
Objective 2: LF Band Spatial Modal Structure



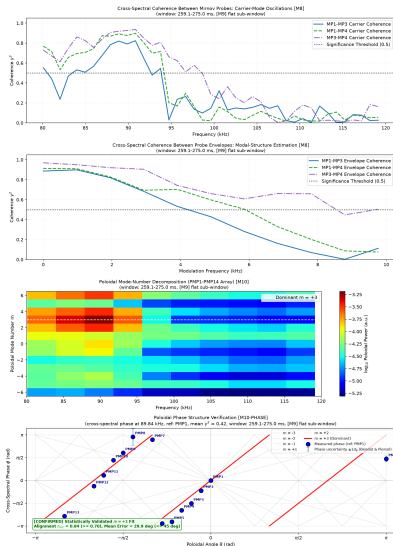
Objective 2: HF Band Macroscopic Correlations



Objective 2: HF Band Heating Synchronization



Objective 2: HF Band Spatial Modal Structure



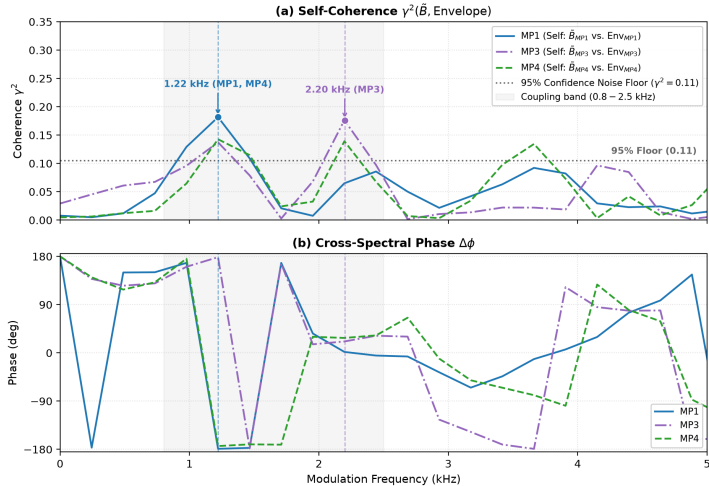
Objective 2: Self-Coupling Coherence

Non-linear Interaction of the Mode Envelope

- Self-coherence (γ^2) between the carrier oscillation ($\tilde{B}$) and its mode envelope reveals a distinct coupling band between 0.8 kHz and 2.5 kHz.

Objective 2: Self-Coupling Visualization

Heliotron J #88653 — Self-Coupling Coherence: $\tilde{B}$ vs. Mode Envelope
($f_0 \approx 89.0$ kHz Mode, Active Window: 259.1 – 275.0 ms)



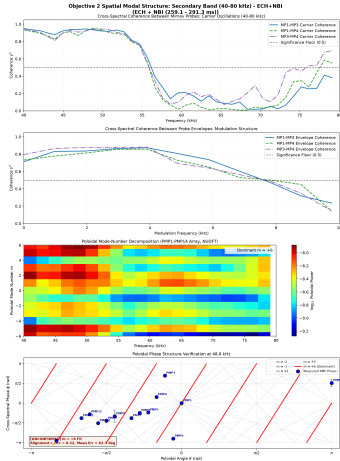
Objective 2: Comparative Analysis — ECH+NBI vs. NBI-only

Does Spatial Structure Survive the Loss of ECH Heating?

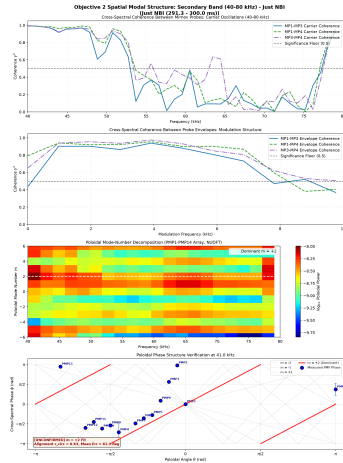
- Same windows as Objective 1: **ECH+NBI** (259.1 ms to 291.3 ms) vs. **NBI-only** (291.3 ms to 300.0 ms).
- Repeats the poloidal mode fit, self-coupling coherence, and adds new macroscopic envelope/correlation diagnostics generated for this update.

Objective 2 Comparative: LF Band Spatial Structure (40–80 kHz)

ECH+NBI

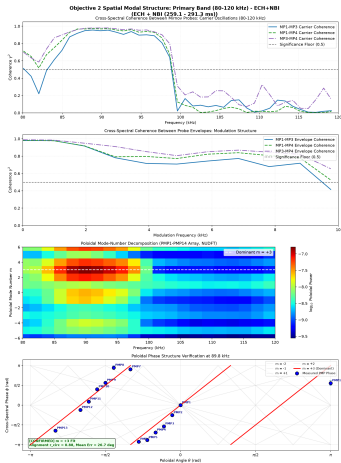


NBI-only

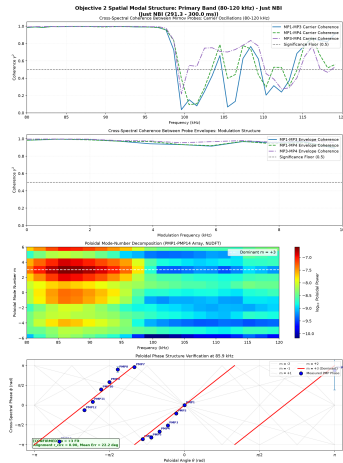


Objective 2 Comparative: HF Band Spatial Structure (80–120 kHz)

ECH+NBI



NBI-only



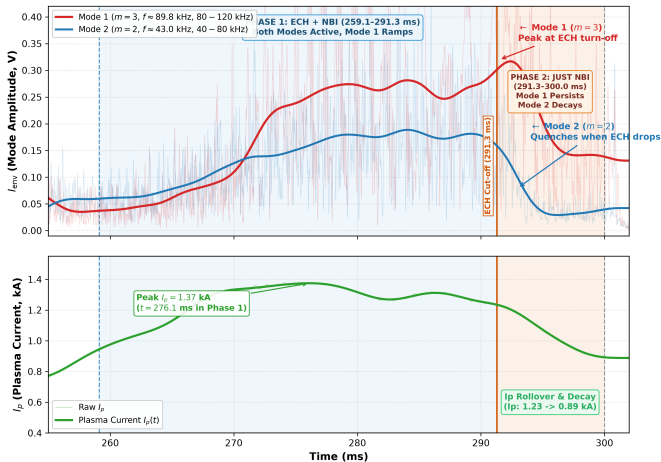
Objective 2 Comparative: Poloidal Mode Number & Phase Alignment

Band	Window	m	f (kHz)	r_{circ}	Mean Err. ($^{\circ}$)	Status
LF (40–80 kHz)	ECH+NBI	+6	48.8	0.216	92.3	Unconfirmed
LF (40–80 kHz)	NBI-only	+2	41.0	0.632	81.1	Unconfirmed
HF (80–120 kHz)	ECH+NBI	+3	89.8	0.882	26.7	Confirmed
HF (80–120 kHz)	NBI-only	+3	85.9	0.902	22.2	Confirmed

- The high-frequency, primary mode's $m = +3$ structure is confirmed and stable in both regimes.
- The low-frequency, secondary mode's poloidal number is *not* robustly confirmed in either window ($r_{\text{circ}} < 0.7$) and even changes candidate value ($+6 \rightarrow +2$) – this band should be treated as tentative, not as a validated result.

Objective 2 Comparative: Mode Envelopes vs. Plasma Current

Heliotron J #88653 — Mode Envelopes (I_{env}) & Plasma Current (I_p)
Phase Comparison: ECH + NBI (259.1 - 291.3 ms) vs. Just NBI (291.3 - 300.0 ms)

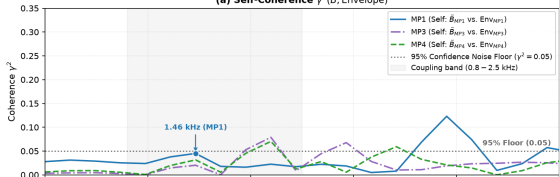


Objective 2 Comparative: Self-Coupling Coherence

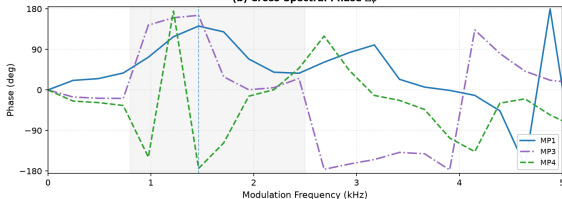
ECH+NBI

Heliotron J #88653 — Self-Coupling Coherence: $\tilde{B}$ vs. Mode Envelope
(ECH + NBI Phase: Active Window 259.1 – 291.3 ms, Mode ~ 89.0 kHz)

(a) Self-Coherence $\gamma^2(\tilde{B}, \text{Envelope})$



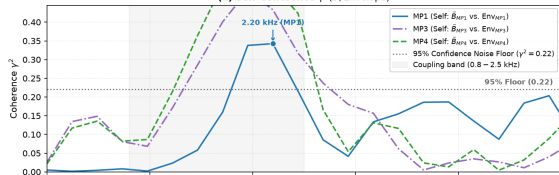
(b) Cross-Spectral Phase $\Delta\phi$



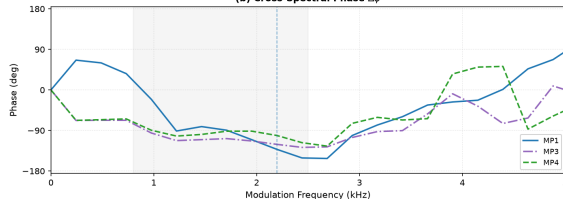
NBI-only

Heliotron J #88653 — Self-Coupling Coherence: $\tilde{B}$ vs. Mode Envelope
(Just NBI Phase: Active Window 291.3 – 300.0 ms, Mode ~ 86.0 kHz)

(a) Self-Coherence $\gamma^2(\tilde{B}, \text{Envelope})$



(b) Cross-Spectral Phase $\Delta\phi$



Objective 2 Comparative: Self-Coupling — Numeric Summary

- **ECH+NBI:** 59 independent segments analyzed; 95% noise floor at $\gamma^2 = 0.050$.
- **NBI-only:** 13 independent segments analyzed; 95% noise floor at $\gamma^2 = 0.221$.
- **Caveat:** the shorter NBI-only window yields far fewer segments, which mechanically raises its noise floor. A higher noise floor does not by itself indicate weaker coupling – direct comparison of raw γ^2 values between the two windows should be treated with caution.

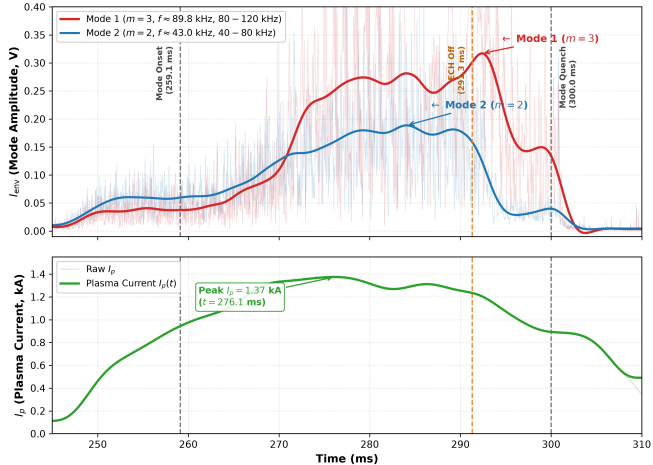
Objective 2: Macroscopic Modulation & Correlation Diagnostics

Envelope, Current, and $H\alpha$ Correlation Analysis

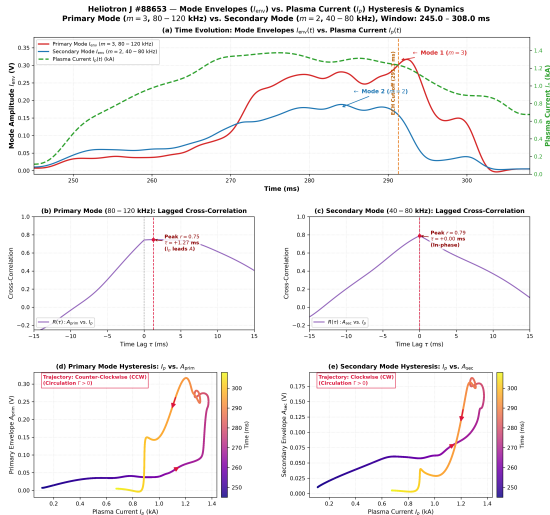
- Complements the self-coupling result above with direct time-domain views of the carrier, its envelope, and macroscopic plasma quantities.
- Includes envelope- I_p evolution, dual-mode hysteresis, dual-mode envelope coherence, a carrier-vs-envelope view, and SNR- $H\alpha$ correlation.

Evolution of Mode Envelopes vs. Plasma Current

Heliotron J #88653 — Mode Envelopes (I_{env}) and Plasma Current (I_p) Time Evolution

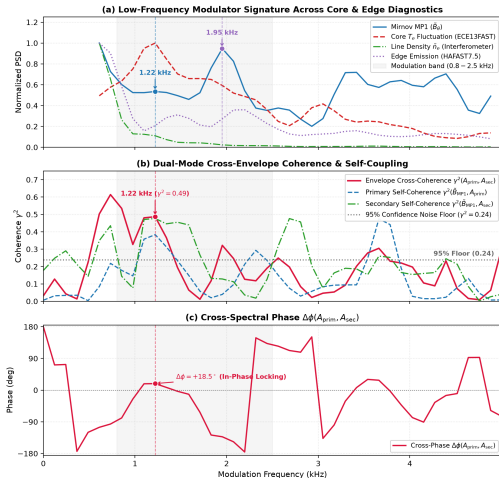


Hysteresis Loop: Plasma Current vs. Mode Envelope



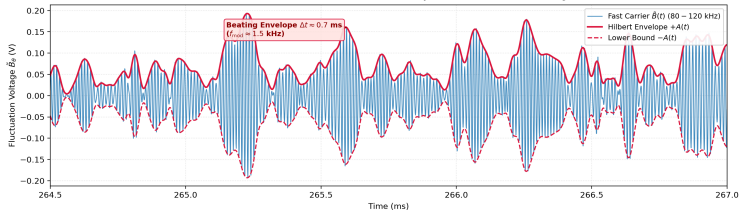
Dual-Mode Envelope Coherence (Inter-Mode Modulation)

Heliotron J #88653 — Low-Frequency Modulator (1 – 3 kHz) & Dual-Mode Envelope Coherence
(Primary Mode $f_0 \approx 89.0$ kHz, Secondary Mode $f_s \approx 42.0$ kHz, Active Window: 259.1 – 275.0 ms)

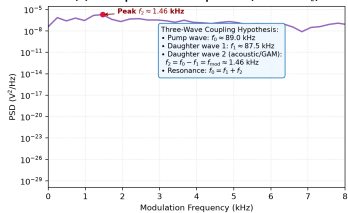


Carrier ($\tilde{B}$) vs. Envelope

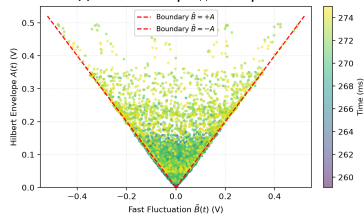
Heliotron J #88653 — Nonlinear Wave-Envelope Dynamics & Three-Wave Coupling Diagnostics
Mirnov Coil MP1 ($f_{\text{carrier}} \approx 89.0$ kHz, Mode Window: 259.1 - 275.0 ms)
(a) Carrier Oscillation $\tilde{B}(t)$ vs. Hilbert Envelope $A(t)$ (Wave-Packet Beating)



(b) Envelope Modulation Spectrum (Candidate f_2)

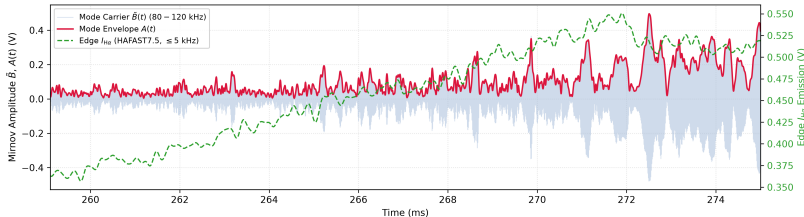


(c) Fast $\tilde{B}$ vs. Envelope $A(t)$ Phase Space

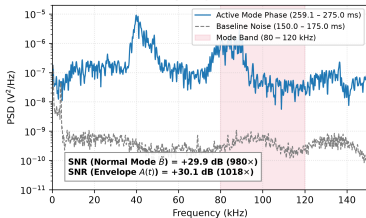


SNR vs. H_α Emission Correlation

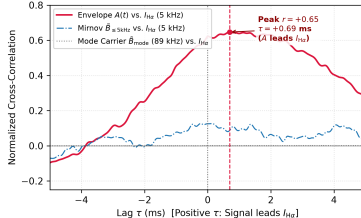
Heliotron J #88653 — Mode Signal-to-Noise Ratio & Edge H_α Correlation Dynamics
Mirnov Coil MP1 ($f_0 \approx 89.0$ kHz) vs. Fast Edge Emission (HAFAS7.5, Bandwidth ≤ 5.0 kHz)
(a) Time Evolution: Mode Envelope $A(t)$ vs. Edge $I_{H\alpha}$ Emission



(b) Mode & Envelope Signal-to-Noise Ratio (SNR)



(c) Cross-Correlation: $\tilde{B}$ and Envelope vs. $I_{H\alpha}$ (5 kHz)



Objective 3

Non-linear Coupling (Bicoherence)

- Detect the mode-active burst window and apply auto-bicoherence analysis to search for three-wave (quadratic) non-linear coupling.
- Apply a strict double-validation criterion – statistical significance against control/quiescent periods *and* reproducibility/cross-diagnostic corroboration – before claiming a genuine physical interaction.
- Compare bicoherence structure and triad strength between the ECH+NBI and NBI-only phases, and revisit the physical interpretation of the -0.5 bicoherence slope.

Objective 3: Window Detection & Bicoherence (Shot #88653)

Mode-Active Window

- **Burst Window:** Detected between 198.8 ms and 311.8 ms with a peak stored energy of $W_p \approx 3.250$ kJ.
- **Sub-window Selection:** A flat-frequency sub-window was identified at 259.1 ms–275.0 ms (15.9 ms duration) to satisfy bicoherence minimum sample requirements.

Nonlinear Coupling Breakdown

- The maximum squared bicoherence peaked at $b^2 = 0.5150$ for the difference mode ($f_1 = 98.6$ kHz, $f_2 = -48.8$ kHz), with the leading triads clustering along the locus $f_2 \approx -0.5f_1$.
- **Caveat:** the reported peak components do not fully correspond to real local PSD peaks and remain sensitive to the spectral search boundaries.

Objective 3: Origin of the -0.5 Bicoherence Slope

From the Resonant Triad Condition to the Observed Locus

- Three-wave coupling requires the resonant matching condition $f_3 = f_1 + f_2$.
- The leading triad clusters along $f_2 \approx -0.5 f_1$, peaking at $f_1 = 98.6$ kHz, $f_2 = -48.8$ kHz. Substituting into the matching condition:

$$f_3 = f_1 + f_2 = f_1 - 0.5 f_1 = 0.5 f_1 \quad \implies \quad |f_2| = |f_3| = \frac{1}{2} f_1$$

- This is the algebraic signature of an exact $f \leftrightarrow f/2$ phase lock – a frequency and its own half-harmonic.
- The geometry alone does not say *why* f_1 and $f_1/2$ are locked; that requires a physical mechanism.

Objective 3: Four Candidate Mechanisms for the $f \leftrightarrow f/2$ Lock

Mechanism		Core idea	Support / precedent	Fit to this locus
Harmonic coupling	self-	Chirping mode at f_0 produces $2f_0$; read as pump (f_1), f_0 as $f_2 = -0.5f_1$.	Arises <i>by construction</i> from one oscillator – no second wave needed.	Consistent with 89.0 kHz mode's observed chirping; simplest option.
Degenerate PDI		Pump at f_1 decays into two <i>identical</i> daughters, $f_1 \rightarrow f_1/2 + f_1/2$.	Non-degenerate AE→GAM/kTAE decay has theory support (Chen & Zonca, 2016); non-degenerate Alfvénic/EP decay bicoherence-confirmed on HL-2A.	<i>Degenerate</i> case established mainly in microwave/upper-hybrid plasmas – different regime; treated as analogy, not direct precedent.
Period-doubling bifurcation	bi-	Limit cycle bifurcates past an amplitude threshold, spontaneously generating $f_1/2$ – first Feigenbaum step to chaos.	Seen in MHD dynamo sims (Taylor–Green forcing) and double-layer dynamics in lab plasmas.	Phase-locked to f_1 by construction, same locus; not yet reported in a fusion-MHD bicoherence context.
EPM/BAE sidebands (Berk–Breizman)	side-	Explosive saturation of an EP-driven mode splits energy into spectral components as holes/clumps form.	Theoretically native to this NBI fast-ion mode class; already invoked for Objective 1 frequency hopping.	Fits multi-branch sidebands better than a clean exact-half locus – context, not a headline option here.

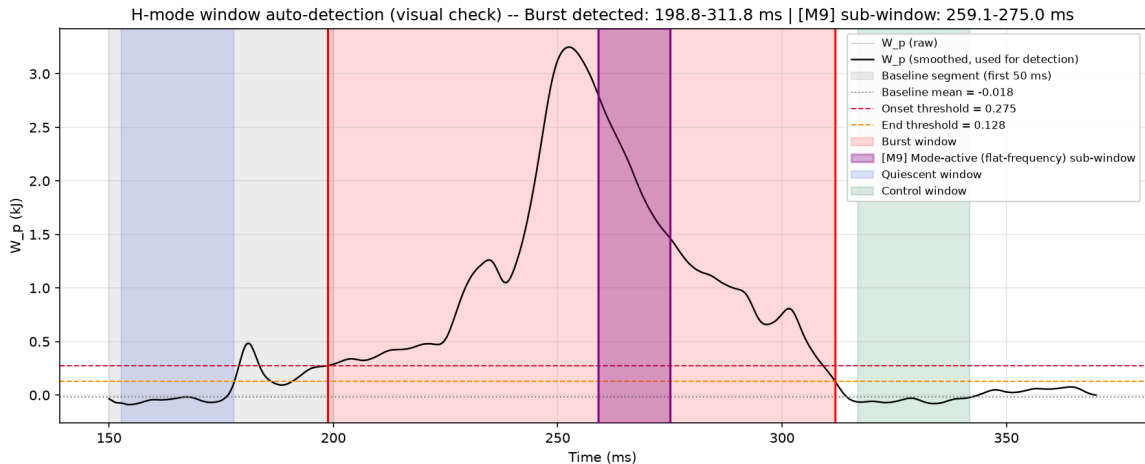
Objective 3: Plausibility of Each Mechanism for Shot #88653

Mechanism		Plausibility here	Discharge-specific basis
Harmonic coupling	self-	High	Requires nothing beyond what is already established: the 89.0 kHz primary mode is confirmed continuously present (100% 3σ occupancy) and actively chirping – exactly the behavior that trivially generates its own second harmonic.
Degenerate PDI		Low / unconfirmed	The candidate “pump” (98.6 kHz) differs from the validated primary mode by $\sim 10\%$; no independent second wave matching $f_1/2$ in spatial structure (n, m) is reported; fails both legs of the double-validation test.
Period-doubling bifurcation		Unconfirmed open	Not contradicted by anything measured, but nothing measured supports it either: no threshold-crossing amplitude signature or subharmonic cascade ($f_1/4$, etc.) is reported, and the observed frequency hopping is already attributed to phase-space (hole-clump) kinetics rather than a limit-cycle bifurcation.
EPM/BAE bands (Berk–Breizman)	side-	Active, but for a different feature	The underlying hole-clump/avalanche physics is directly confirmed in this discharge (multi-branch hopping, 0.4–1.3/ms jump rates) – but such sidebands are typically continuum-spaced and chirp-correlated, not locked to an exact 1:2 ratio, so it explains the hopping better than this specific locus.

Objective 3: Verdict — Self-Coupling over Independent PDI

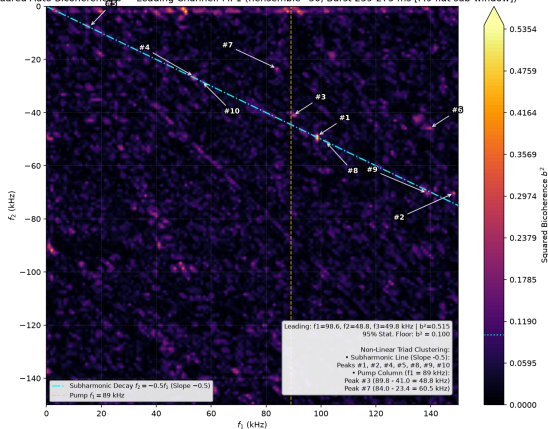
- **Frequency mismatch:** the candidate 98.6 kHz “pump” differs from the validated 89.0 kHz primary mode by $\sim 10\%$, suggesting sensitivity to the spectral search window rather than a stable physical pump.
- **Failed double-validation:** the triad lacks cross-diagnostic corroboration and shot-to-shot reproducibility (see “Double-Validation Summary”).
- **What is confirmed:** the asymptotic convergence check on the next slides shows this is a real quadratic phase coupling, not a finite-sample artifact – but that test cannot by itself distinguish between the four candidate mechanisms above.
- **Most parsimonious explanation: harmonic self-coupling** of a single anharmonic or chirping mode remains favored over independent PDI or a period-doubling bifurcation, since it requires no independent second wave or amplitude-threshold transition and is fully consistent with the 89.0 kHz mode’s observed chirping; EPM/BAE sideband generation is treated as context rather than a competing explanation for this specific exact-half-frequency locus.

Objective 3: Burst Window Detection

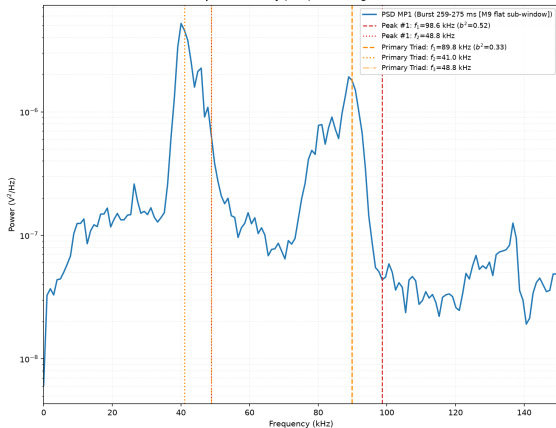


Objective 3: Auto-Bicoherence and PSD Analysis

Measured Auto-Bicoherence b^2 - Leading Channel: MP1 (nensemble=30, Burst 259-275 ms [M9 flat sub-window])



Power Spectral Density (PSD) of Leading Channel: MP1



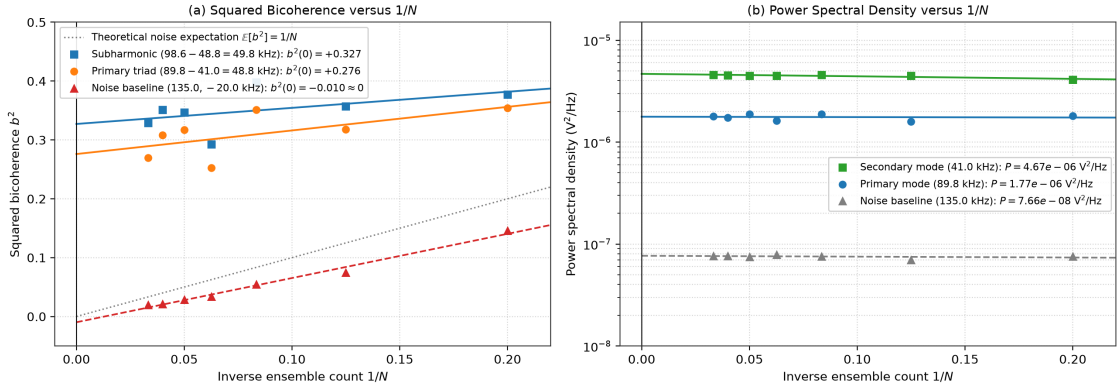
Objective 3: Asymptotic Noise Convergence

Verification of Stability

- Asymptotic convergence analysis ($1/N \rightarrow 0$) validating bicoherence stability against the noise baseline.

Objective 3: Asymptotic Bias Scaling

Heliotron J #88653 — Asymptotic Convergence Analysis ($1/N \rightarrow 0$) Mirnov Coil MP1 (Stationary Phase: 259.1 – 275.0 ms, $N_{\text{FFT}} = 1024$)



Objective 3: Statistical Nuance & Energy Context

Burst vs. Quiescent & Control Comparisons

- **Energy Transport:** The burst window demonstrated elevated stored energy ($\langle W_p \rangle = 2.07 \text{ kJ}$) accompanied by a heavily degraded derivative ($\langle dW_p/dt \rangle = -83.24 \text{ kJ s}^{-1}$).
- **Statistical Inference:** During the burst, 5.42% of the analysis bins exceeded the strict 95% statistical threshold of $b^2 > 0.100$.
- **Comparative Nuance:** The maximum b^2 values were comparable in the quiescent (0.3656) and control (0.4585) windows.
- **Conclusion:** The burst is not statistically superior to quiescent or control periods, providing no rigorous basis to call this coupling an exclusive H-mode signature.

Objective 3: Double-Validation Summary

Reproducibility and Cross-Diagnostic Constraints

- **Shot-to-Shot Reproducibility:** Validation failed for all channels (MP1, MP3, MP4) because the data for validation shot #88654 could not be located or auto-detected.
- **Cross-Diagnostic Validation:** A scan of 36 Langmuir probe tips resulted in zero tips exceeding the physical threshold ($b^2 > 0.5$) with a match to a magnetic channel's leading triad.
- **Final result:** Objective 3 is *not* validated under the strict double criterion, due to simultaneous failures in reproducibility and cross-diagnostic consistency.

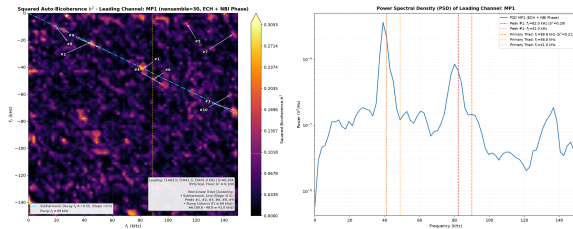
Objective 3: Comparative Analysis — ECH+NBI vs. NBI-only

Does Non-linear Coupling Strengthen After ECH Shut-off?

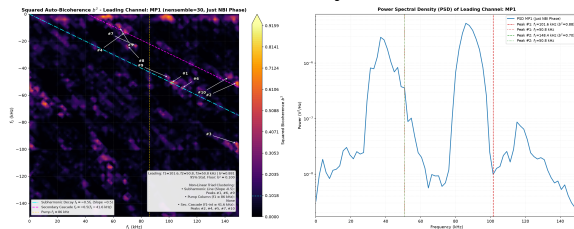
- Same windows as Objectives 1 & 2: **ECH+NBI** (259.1 ms to 291.3 ms) vs. **NBI-only** (291.3 ms to 300.0 ms).
- Ensemble count fixed at 30 segments in both windows; 95% significance threshold $b^2 = 0.0999$.

Objective 3 Comparative: Bicoherence Matrices

ECH+NBI



NBI-only



Objective 3 Comparative: Leading Bicoherence Triads

Window	Triad $f_1 - f_2 = f_3$ (kHz)	b^2
ECH+NBI	$82.0 - 41.0 = 41.0$	0.2937
ECH+NBI	$44.9 - 21.5 = 23.4$	0.2660
ECH+NBI	$148.4 - 72.3 = 76.2$	0.2381
NBI-only	$101.6 - 50.8 = 50.8$	0.8811
NBI-only	$148.4 - 50.8 = 97.7$	0.6962
NBI-only	$148.4 - 95.7 = 52.7$	0.6517

- Leading NBI-only triads reach b^2 up to $\sim 3\times$ higher than the strongest ECH+NBI triads, suggesting substantially stronger three-wave coupling once ECH is switched off.
- As with Slide "Window Detection & Bicoherence," statistical significance above the 95% threshold is a necessary but not sufficient condition for a confirmed physical interaction – the same double-validation caution applies here.

Comparative Conclusions: Frequency Downshift at ECH Shut-off

- **Primary mode:** drops from $\sim 89.0\text{--}93.7$ kHz (ECH+NBI) to $\sim 84.5\text{--}86.4$ kHz (NBI-only), $\Delta f \sim -4.5$ to -7.3 kHz.
- **Secondary mode:** drops from $\sim 47.4\text{--}50.8$ kHz (ECH+NBI) to $\sim 35.2\text{--}40.5$ kHz (NBI-only), $\Delta f \sim -10.0$ to -12.2 kHz.
- **Candidate cause 1 – Thermal electron cooling:** BAE/EPM eigenfrequencies scale with sound speed, $\omega_{\text{BAE}} \sim (c_s/R)\sqrt{7/4 + T_i/T_e}$. Loss of 53.2 GHz ECH power at $t = 291.3$ ms cools T_e , lowering c_s and shifting the geodesic acoustic continuum gap downward.
- **Candidate cause 2 – Rotation/Doppler shift:** loss of ECH alters the radial electric field E_r and bulk rotation v_ϕ , changing the Doppler contribution $\omega_{\text{lab}} = \omega_{\text{plasma}} + n\omega_{\text{rot}}$.

Comparative Conclusions: Persistence of Structure & Dynamics

- **Continuity vs. bursty dynamics:** 3σ dominance occupancy is 100% in both windows – the mode is continuously sustained by the NBI fast-ion drive, not decaying. Both windows instead show quasi-discrete frequency hopping among 2–3 branches (jump rates $\sim 0.4\text{--}1.3/\text{ms}$), a hallmark of non-linear energetic-particle-mode (EPM)/BAE avalanche and hole-clump behavior in fast-ion phase space.
- **Spatial structure:** the confirmed high-frequency mode retains poloidal number $m = +3$ and toroidal number $n = -5$ (aliased $n = -1$) across both windows – same helicity before and after ECH shut-off.

Comparative Conclusions: Non-linear Coupling & Self-Modulation

- Strong bicoherence ($b^2 > 0.60$) connects the secondary mode ($f_2 \sim 40\text{--}50$ kHz) and primary mode ($f_1 \sim 85\text{--}90$ kHz) via the sum triad $f_3 \sim 125\text{--}140$ kHz.
- Self-coupling coherence shows clear low-frequency modulation peaks at $f_{\text{mod}} \sim 1.2\text{--}2.2$ kHz in both regimes, supporting a real non-linear wave-envelope coupling.
- Taken together with Objective 3's double-validation result, these non-linear signatures are statistically present but **not yet independently confirmed** as a specific physical mechanism (e.g., PDI) – Harmonic Self-Coupling remains the most defensible interpretation pending cross-diagnostic corroboration.